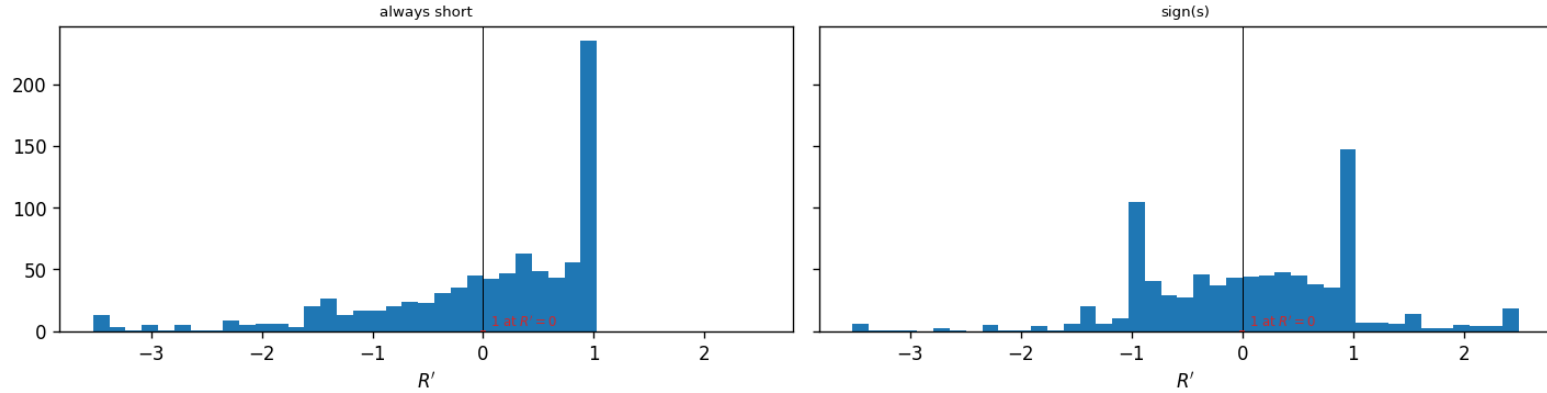


ODTE 15:30–16:00 nearest-OTM straddle: per-rule return summary, models compared on the same 866 days

	n	mean	std	min	25%	50%	75%	max	skew	ex. kurt	t	Sharpe _{ann}	n_{buy}	%buy
<i>Short volatility (always short)</i>														
all models (no forecast)	866	0.014	1.128	-10.32	-0.449	0.327	0.928	1.00	-2.31	10.89	0.38	0.20	0	0.0
<i>sign(s) (± 1 on the sign of s)</i>														
baseline (HAR + calendar OLS)	866	0.069	1.126	-5.42	-0.643	0.159	0.882	10.32	0.58	10.71	1.79	0.97	286	33.0
block-diagonal ridge	866	0.095	1.124	-5.42	-0.717	0.130	0.870	10.32	0.92	10.61	2.48	1.34	346	40.0
block-diagonal ridge, without the FOMC columns	866	0.090	1.124	-5.42	-0.733	0.116	0.849	10.32	1.00	10.59	2.36	1.27	355	41.0
LightGBM [†]	866	0.103	1.123	-5.42	-0.642	0.161	0.893	10.32	1.01	10.58	2.71	1.46	290	33.5
XGBoost [†]	866	0.106	1.123	-5.42	-0.627	0.199	0.908	10.32	0.91	10.62	2.78	1.50	279	32.2
lasso (causally tuned) [†]	866	0.100	1.123	-6.84	-0.653	0.146	0.913	10.32	0.46	10.78	2.62	1.41	321	37.1
lasso ($\alpha = 10^{-4}$)	866	0.107	1.123	-5.42	-0.729	0.116	0.855	10.32	1.22	10.50	2.80	1.51	358	41.3
elastic net (causally tuned) [†]	866	0.072	1.125	-6.84	-0.683	0.130	0.883	10.32	0.46	10.74	1.90	1.02	307	35.5
<i>per-bar forecasts: one regression for the 15:30–16:00 bar alone</i>														
per-bar ridge (HAR + calendar)	866	0.088	1.124	-4.18	-0.639	0.122	0.894	10.32	0.99	10.59	2.32	1.25	327	37.8
per-bar ridge (all features)	866	0.125	1.121	-5.42	-0.657	0.177	0.893	10.32	1.13	10.53	3.29	1.77	316	36.5
per-bar lasso (all features)	866	0.102	1.123	-5.42	-0.673	0.160	0.874	10.32	1.00	10.58	2.67	1.44	330	38.1
per-bar elastic net (all features)	866	0.112	1.122	-5.42	-0.653	0.167	0.882	10.32	0.99	10.59	2.92	1.58	336	38.8
per-bar ridge (live-feasible set)	866	0.119	1.121	-5.42	-0.653	0.154	0.907	10.32	1.00	10.59	3.12	1.68	348	40.2
per-bar lasso (live-feasible set)	866	0.111	1.122	-5.42	-0.652	0.161	0.862	10.32	0.98	10.59	2.92	1.58	332	38.3
per-bar elastic net (live-feasible set)	866	0.120	1.121	-5.42	-0.633	0.172	0.883	10.32	0.93	10.62	3.15	1.70	333	38.5

block-diagonal ridge, midpoint fills: daily return of the position, 1st–99th percentile window (bars: the days with $R' \neq 0$; the red bar at 0 is the rest)



Distribution of the daily return of the position taken, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the straddle expires worthless).

Straddle here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). $n_{\text{buy}}/\% \text{buy}$ count days the rule is net long ($q_t > 0$). The short-volatility rule takes no forecast, so it is one row for all models. Sharpe_{ann} = (mean/std) $\times \sqrt{252}$; every other column is daily.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending.

Per-bar forecasts (the rows under their own subheading): a separate regression for the 15:30–16:00 bar alone. Each day it is trained on that half hour of the previous 2,000 sessions and nothing else; the target is the bar’s realized variance over its time-of-day baseline, square-rooted; the inputs, all known at 15:30, are each series’ means over the last 1, 5, 25, 125, 625 and 3,125 half-hour bars of the full 24-hour series, is-present flags and calendar dummies. Ridge penalty chosen from $\{0.01, \dots, 1000\}$ every 250 sessions from past data only (fit on the earlier part of the window, scored on its last 125 sessions); coefficients re-solved every session; the lasso and elastic net differ only in the penalty. Column sets: HAR + calendar only; all features; the live-feasible set (16 series a 15:30 forecaster can rebuild: ES return moments, volume and trade count, VIX, VVIX, VIX3M, the FOMC calendar). The prediction is turned into a variance by the same recalibration as every other row.